

Multiple Choice (10 marks)

1. As of 2020, which architecture is best for classifying high-resolution images?
 - (a) convolutional networks
 - (b) graph networks
 - (c) fully connected networks
 - (d) RBF networks
2. MLE estimates are often undesirable because
 - (a) they are biased
 - (b) they have high variance
 - (c) they are not consistent estimators
 - (d) None of the above
3. K-fold cross-validation is
 - (a) linear in K
 - (b) quadratic in K
 - (c) cubic in K
 - (d) exponential in K
4. Which one of the following is the main reason for pruning a Decision Tree?
 - (a) To save computing time during testing
 - (b) To save space for storing the Decision Tree
 - (c) To make the training set error smaller
 - (d) To avoid overfitting the training set
5. Which of the following is NOT supervised learning?
 - (a) PCA
 - (b) Decision Tree
 - (c) Linear Regression
 - (d) Naive Bayesian

True / False (10 marks)

Answer True or False

6. True or False: A larger number of test examples is required to achieve statistically significant results when the error rate is smaller.
7. True or False: In PCA, transforming data to zero mean is essential to achieve the same projection as Singular Value Decomposition (SVD).
8. True or False: ImageNet comprises a diverse collection of images with varying resolutions, contributing to its robustness in image classification tasks.
9. True or False: The null space of the matrix A has a dimensionality of 2, indicating there are two free variables in the system.
10. True or False: Both Bayesians and frequentists agree on the necessity of using prior distributions in all probabilistic models.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Write a function to select the n th items of a list.
12. Write a function to find numbers within a given range where every number is divisible by every digit it contains.

End of Paper